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Grid computing is a distributed architecture that coordinates various loosely coupled computers to act as a virtual supercomputer for a single, massive task, whereas cloud computing provides on-demand access to a shared pool of configurable resources (like servers and storage) as a service. While grid computing is designed for high-performance research and specific compute-intensive projects, cloud computing focuses on general-purpose business agility and delivering seamless web services. In terms of scalability, grids are often limited by the physical hardware of the participating nodes, while clouds offer "elasticity," allowing resources to scale up or down instantly based on user demand. Real-world applications of grid computing include complex scientific simulations like earthquake modeling, whereas cloud computing powers everyday platforms like Netflix, Gmail, and e-commerce websites.